MATH MA’AM ACADEMY

Where Every Subject Clicks.

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STANDARD VI • SCIENCE

UNIT 5: WATER & HYDROGEN

Structured revision notes based on the supplied textbook pages, with essential clarifications for exam preparation.

QUICK FACTS

Fact	Key information
Ocean water	About 97.2% of Earth's water (textbook figure).
Water formula	H_2O; hydrogen : oxygen atoms = 2 : 1.
Mass ratio in water	Hydrogen : oxygen = 1 : 8.
Maximum density	Approximately 1 g/cm^3 at 4°C.
Freezing / boiling	0°C / 100°C at standard atmospheric pressure.
Hardness-causing ions	Mainly calcium (Ca^{2+}) and magnesium (Mg^{2+}).
Hydrogen test	Burns with a characteristic 'pop' sound in a controlled test.

Water-distribution figures can vary slightly by source. For textbook-based exam questions, learn the figures printed in your prescribed book.

1. THE HYDROSPHERE

The hydrosphere includes all water on Earth: oceans, seas, rivers, lakes, glaciers, groundwater and water vapour in the atmosphere. Oceans form the largest visible part of the hydrosphere.

Distribution of water on Earth

Reservoir	Textbook estimate	Importance / note
Oceans	97.2%	Largest reservoir; mainly saline water.
Polar caps and ice sheets	2.15%	Store freshwater in frozen form.
Rivers and lakes	0.0171%	A small fraction of total water; important for people and ecosystems.
Atmosphere	0.001%	Water vapour supports the water cycle and helps prevent desiccation.

- Hydrology: study of water, its distribution, movement and use.
- Water cycle: circulation of water through evaporation, condensation, precipitation, runoff and infiltration.
- Only a small fraction of Earth's total water is readily accessible freshwater; most is saline or locked in ice.

Importance of the hydrosphere

- Helps regulate Earth's temperature and climate.
- Provides water for drinking, agriculture, industry and domestic activities.
- Supports aquatic organisms and terrestrial life.
- Transfers water between oceans, land and the atmosphere through the water cycle.

Fresh water and salt water

- Fresh water: commonly found in rivers, many lakes, glaciers and groundwater; dissolved substances depend on surrounding rocks and soil.
- Salt water: mainly found in seas and oceans. It contains sodium chloride and smaller amounts of other dissolved salts, including compounds of magnesium, calcium and potassium.
- Seawater is generally unsuitable for drinking or irrigation without treatment because of its high salt content.
- Desalination can produce freshwater from seawater. The natural water cycle also replenishes freshwater supplies.

The textbook percentages are rounded estimates and may not add to exactly 100%.

2. COMPOSITION OF WATER

Water is a compound of hydrogen and oxygen. Henry Cavendish's work in 1781 is associated with demonstrating water's composition; Antoine Lavoisier later helped establish its chemical nature and naming.

Property	Key fact
Molecular formula	H_2O
Atoms in one molecule	2 hydrogen atoms and 1 oxygen atom
Atom ratio (H : O)	2 : 1
Mass ratio (H : O)	1 : 8
Electrolysis equation	$2H_2O \rightarrow 2H_2 + O_2$
Gas volume ratio (H_2 : O_2)	2 : 1 under the same conditions

Electrolysis of water

- Electric current decomposes water into hydrogen and oxygen when suitable apparatus and conditions are used.
- In the usual setup, hydrogen forms at the cathode (negative electrode) and oxygen at the anode (positive electrode).
- Twice as much hydrogen as oxygen is produced by volume under the same conditions.
- The mass ratio follows from relative atomic masses: 2 g of hydrogen combine with 16 g of oxygen.

Illustrative fact

In 18 g of water, hydrogen contributes 2 g and oxygen contributes 16 g. Complete electrolysis theoretically yields 2 g of hydrogen and 16 g of oxygen, assuming no losses.

3. PHYSICAL PROPERTIES OF WATER

- Pure water is transparent, colourless, odourless and tasteless.
- At standard atmospheric pressure, pure water freezes at 0°C and boils at 100°C.
- Dissolved non-volatile substances generally raise water's boiling point; atmospheric pressure also affects it.
- At higher altitudes, lower atmospheric pressure lowers the boiling point, so food may take longer to cook.
- Water has a maximum density of approximately 1 g/cm³ at 4°C.
- Ice is less dense than liquid water and floats, helping insulate water below and supporting aquatic life in cold climates.
- Pure water is a poor electrical conductor. Dissolved salts produce mobile ions, making the solution more conductive.
- Water expands as it cools below 4°C and when it freezes. This is called the anomalous behaviour of water.

4. HARD WATER AND SOFT WATER

Feature	Soft water	Hard water
Action with soap	Lathers readily	Less lather; may form scum
Hardness-causing ions	Low concentration	Often calcium and magnesium ions
Practical effects	Useful for washing	Can waste soap and leave deposits

Causes of hardness

- Natural water dissolves minerals while flowing over or through soil and rocks.
- Calcium and magnesium salts—especially bicarbonates, chlorides and sulphates—can cause hardness.
- Temporary hardness: commonly caused by calcium and magnesium bicarbonates; it can be removed by boiling.
- Permanent hardness: commonly caused by calcium and magnesium chlorides and sulphates; boiling alone does not remove it effectively.

Disadvantages of hard water

- Consumes more soap because some soap forms insoluble scum.
- May leave deposits on utensils, pipes and heating equipment.
- Can be unsuitable for some industrial processes.

Methods of softening

- Boiling: removes temporary hardness by forming precipitates from certain dissolved compounds.
- Washing soda (Na_2CO_3): carbonate ions react with calcium or magnesium ions to form insoluble carbonates.
- Ion exchange / permutit process: exchanges hardness-causing ions for other ions, commonly sodium ions.

Practical 1 — Softening hard water with washing soda

- Add a small quantity of washing soda to hard water. Insoluble calcium carbonate and/or magnesium carbonate may form.
- Filter the mixture to remove the precipitate.
- Test the filtrate with soap solution. Improved lather indicates reduced hardness.
- Role of washing soda: supplies carbonate ions that remove calcium and magnesium ions as insoluble precipitates.

Practical 3 — Comparing dissolved sulphates

- Prepare equal volumes of water containing sodium sulphate, ammonium sulphate, calcium sulphate and magnesium sulphate separately.
- Add equal amounts of soap solution and shake each sample consistently.
- Calcium and magnesium ions cause hardness and reduce lather; sodium and ammonium ions do not cause hardness in the same way.

5. WATER POLLUTION

Water pollution is contamination by suspended particles, dissolved chemicals, sewage, industrial waste, excessive nutrients, algae or disease-causing microorganisms, making water harmful or unsuitable for use.

Major sources

- Industrial waste: chemicals and heavy metals discharged from factories.
- Sewage: untreated domestic wastewater and human excreta entering water bodies.
- Synthetic detergents: excessive use of persistent, poorly biodegradable detergents.
- Agrochemicals: fertilizers and pesticides washed from fields into streams, rivers and lakes.

Harmful effects

- May spread water-borne diseases such as typhoid, cholera and dysentery.
- Polluted water may contain harmful bacteria, protozoa and parasitic worms.
- Pollution can damage fish and other aquatic organisms; excess nutrients may cause algal blooms and oxygen depletion.
- Heavy metals may cause poisoning; lead and mercury are examples of toxic heavy metals.

Prevention

- Treat industrial effluents to remove or neutralize harmful substances before discharge.
- Treat sewage to reduce organic matter and pathogens before release or reuse.
- Avoid excessive fertilizer and pesticide use and prevent contaminated runoff.
- Dispose of garbage and chemicals responsibly; protect water sources from contamination.

6. PURIFICATION OF WATER

Drinking water should be free from harmful suspended impurities and disease-causing organisms. Some dissolved chemicals may also need removal, depending on the source.

Method	Main purpose / limitation
Sedimentation	Allows heavier suspended particles to settle.
Filtration	Removes many suspended particles; ordinary filters do not remove every dissolved substance or pathogen.
Boiling	Kills many disease-causing microorganisms when done correctly; does not remove dissolved salts or chemical pollutants.
Chlorination	Disinfects water when the correct dose and contact time are used.
Ozonation	Uses ozone to disinfect water; requires suitable equipment.
Aeration	Exposes water to air; can remove some volatile substances and improve taste/odour, but is not complete disinfection.
Potassium permanganate	An oxidant used in some treatment contexts; not a universal substitute for validated drinking-water disinfection.

Remember: sedimentation and filtration mainly remove suspended matter. Disinfection targets microorganisms. Clear-looking water is not necessarily safe to drink.

7. HYDROGEN

Hydrogen is the lightest element and the most abundant element in the universe. The Sun and stars consist largely of hydrogen. On Earth, hydrogen is commonly combined with other elements in water and compounds such as hydrocarbons, sugars, fats and proteins.

Laboratory preparation

Hydrogen can be prepared by reacting zinc or magnesium with a suitable dilute acid, such as dilute hydrochloric acid.

Reaction	Balanced equation
Zinc + hydrochloric acid	$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$
Magnesium + hydrochloric acid	$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$

- Bubbles of hydrogen form as the metal reacts with dilute acid.
- Hydrogen is highly flammable and must be handled only with appropriate laboratory precautions.

Properties of hydrogen

Test / property	Observation or inference
Colour	Colourless
Odour	Odourless
Effect on litmus	No change to red or blue litmus; neutral in this test
Solubility	Sparingly soluble in water
Density	Much lighter than air
Burning test	Characteristic 'pop' sound in a carefully controlled test
Flame	A pale blue flame may be observed

Practical 2 — Studying hydrogen gas

- Set up the apparatus as directed by the teacher and add the prescribed dilute acid to the metal.
- Observe gas bubbles and test the gas with red and blue litmus papers.
- A characteristic 'pop' sound on a carefully controlled test supports the identification of hydrogen.

Safety: Use dilute acid, not concentrated hydrochloric acid, for a routine student demonstration. Never bring a flame near a gas-generation apparatus while gas is still being produced. The teacher must control the setup, ventilation and flame test.

Uses of hydrogen

- Oxy-hydrogen flame: a hydrogen–oxygen flame reaches high temperatures and has been used for cutting and welding.
- Fuel: hydrogen can be used as a fuel; liquid hydrogen is used as rocket propellant in some launch systems.
- Hydrogenation: hydrogen is added to unsaturated compounds. Industrial hydrogenation of vegetable oils can increase saturation and change oil properties.

8. EXAM-READY TERMS AND COMPARISONS

Term	Meaning
Hydrosphere	All water present on, under and above Earth's surface.
Hydrology	Study of water, including its distribution and movement.
Electrolysis	Chemical decomposition caused by an electric current.
Anomalous behaviour of water	Expansion below 4°C and on freezing.
Hard water	Water that does not readily form lather because of dissolved ions, especially calcium and magnesium.
Softening	Reducing or removing hardness-causing ions from water.
Water pollution	Contamination that harms water quality or makes it unsafe or unsuitable.
Disinfection	Treatment intended to destroy or inactivate disease-causing microorganisms.
Hydrogen pop test	Characteristic sound produced when hydrogen ignites in a small, controlled test.

Revise the definitions, observations, ratios, equations, uses and scientific reasons in each section.